EXCEL DASHBOARD · PROJECT GUIDE

StarTechOne student enrolments dashboard

How the workbook was built, what each page shows, what the data says, and how to use and maintain the dashboard.



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Workbook: StarTechOne_Student_Enrolments_Redesigned.xlsx (2 dashboard pages, 2 analysis sheets, data and dictionary)
Data: 560 synthetic student records, October 2022 to September 2026, as at 23 October 2026
Tools: Microsoft Excel: Tables, PivotTables, slicers, GETPIVOTDATA, charts
Version: October 2026

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Who this guide is for. Managers who will read the dashboard (sections 1, 7, 9 and 10) and analysts who will maintain or extend the workbook (sections 3 to 6, 11 and 12).

About the data. The dataset is synthetic and was created for portfolio use. It contains no real student records; the findings illustrate the method, not a real school.

1 · EXECUTIVE SUMMARY

Intake nearly doubled; online learners are the ones who drop out

560
students enrolled, +88% from 2022/23 to 2025/26

87%
completion (459 of 529 whose course ended)

68%
online completion, against 94% in class

91%
of ₦60.5M fees collected

Area	What the data shows
Growth	Intake rose 97, 122, 159, 182 across the four academic years; Web Development grew every year (44, 52, 58, 71).
Student mix	SIWES and IT placement students are 38% of everyone trained; 55% of students are under 25; 40% are women.
Completion	87% overall; online 68% vs 94% in class. Regular students, the only ones who learn online, complete at 81%.
Attendance	Completers attended 88% of classes; students who dropped out attended 50%.
Results	Average score 73.1; 64% of completers earned an A; 457 of 459 completers were certified; average rating 4.3 of 5.
Fees	₦54.7M of ₦60.5M paid; ₦5.7M owed, of which ₦3.3M sits with students who did not complete.
Certificates	40 of 457 certificates not yet collected; 19 have waited more than three months.

2 · THE BRIEF

One place to see intake, outcomes and fees

StarTechOne is a (fictional) tech training academy in Nigeria teaching Data Analysis and Web Development to paying learners and to university (SIWES) and polytechnic (IT) students on industrial placements. Management needed one workbook that answers the questions a school manager actually asks:

- Who joins us, when, for which course, and how did they hear about us?
- Who finishes, how well do they do, and are certificates being collected?
- How much have we charged, collected, and still not received?

Objectives

1. Show enrolment growth and the make-up of the student body.
2. Measure completion, scores, certification and satisfaction.
3. Track fees charged, paid and outstanding.
4. Let a non-technical user slice every number with three clicks and no formulas.
5. Surface a short list of actions backed by the numbers.

The whole solution lives in one `.xlsx` file: no VBA, no Power Query and no external connection, so it filters and recalculates inside Excel alone.

3 · WORKBOOK STRUCTURE

Eight sheets: two dashboards on top, the engine hidden underneath

Sheet	Visibility	Purpose
Enrolment Dashboard	Visible	Page 1: intake, demographics, tracks, geography, acquisition channels
Outcomes Dashboard	Visible	Page 2: completion, scores, grades, certificates, ratings, fees
Data	Visible	560 rows × 30 columns in the Excel Table `tblEnrolments`
Data Dictionary	Visible	Definition of every column
Enrolment Analysis	Visible	Supporting tables and written takeaways for page 1
Outcomes Analysis	Visible	Supporting tables and written takeaways for page 2
Calc	Hidden	14 PivotTables (`calc_*`) that respond to the slicers
Model	Hidden	GETPIVOTDATA formula layer that feeds every chart and KPI

One Excel Table of 560 students and 30 columns

Group	Columns
Identity and enrolment	StudentID, EnrolmentDate, Programme, Track, StudentType, Status
Demographics	Gender, Age, State
Delivery	LearningMode, DurationWeeks, CourseEndDate
Money	FeeNGN, AmountPaidNGN, PaymentStatus
Acquisition	HeardAboutUs
Performance	Attendance, Completed, FinalScore, Grade, Rating
Certification	Certified, CertificateNo, CertificateDate, CertificateCollected

Calculated columns (structured references, so new rows fill in automatically)

Column	Logic
`AcademicYear`	October to December belong to the year starting that calendar year (Oct 2024 → 2024/25); January to September belong to the year that started the previous October
`EnrolMonth`	`TEXT(EnrolmentDate, "mmm")` for the seasonality chart
`AgeGroup`	18 to 19, 20 to 24, 25 to 29, 30 to 34, 35 and above
`OutstandingNGN`	`FeeNGN - AmountPaidNGN`
`CertificateYear`	Year of CertificateDate, or "Not issued"

Business rules in the data

- **Student types:** Regular are paying learners; SIWES are university students on industrial work experience; IT are polytechnic students on industrial training.
- **Learning mode:** only Regular students can learn online.
- **Completed:** Yes, No (dropped out) or In progress; in-progress students are left out of completion rates.
- **Grades:** A = 70+, B = 60 to 69, C = 50 to 59, F = below 50. Certified = completed and scored 50 or more.

Student	Enrolment	Programme	Track	StudentT	Status	Gende	Ag	State	LearningM	DurationWi	FeeNGN	AmountPaid	PaymentSt	HeardAbou
STO0001	05 Oct 2022	Web Development	Full Stack Web Development	Regular	Graduate	Male	25	FCT Abuja	Online	20	220,000	220,000	Paid in full	WhatsApp
STO0002	06 Oct 2022	Data Analysis	Excel and Power BI	Regular	Working professional	Female	26	Niger	Physical	8	80,000	80,000	Paid in full	WhatsApp
STO0003	11 Oct 2022	Data Analysis	Full Data Analysis (Excel, SQL, Power BI)	Regular	Graduate	Male	25	FCT Abuja	Online	16	180,000	120,000	Part paid	Instagram
STO0004	12 Oct 2022	Data Analysis	Python for Data Analysis	Regular	Graduate	Male	27	FCT Abuja	Online	10	120,000	120,000	Paid in full	Friend or fam
STO0005	12 Oct 2022	Data Analysis	SQL for Data Analysis	SIWES	Undergraduate (University)	Female	24	FCT Abuja	Physical	12	70,000	70,000	Paid in full	School place
STO0006	18 Oct 2022	Data Analysis	SPSS for Research	IT	Student (Polytechnic)	Male	20	Niger	Physical	12	70,000	70,000	Paid in full	School place
STO0007	23 Oct 2022	Data Analysis	Python for Data Analysis	Regular	Working professional	Male	22	Nasarawa	Physical	10	120,000	120,000	Paid in full	X (Twitter)
STO0008	23 Oct 2022	Web Development	Full Stack Web Development	Regular	Working professional	Female	25	FCT Abuja	Physical	20	220,000	220,000	Paid in full	Instagram
STO0009	26 Oct 2022	Data Analysis	Excel and Power BI	Regular	Working professional	Male	18	FCT Abuja	Physical	8	80,000	80,000	Paid in full	Instagram
STO0010	08 Nov 2022	Data Analysis	Excel and Power BI	Regular	Working professional	Male	27	FCT Abuja	Online	8	80,000	0	Not paid	WhatsApp
STO0011	11 Nov 2022	Data Analysis	Full Data Analysis (Excel, SQL, Power BI)	Regular	Graduate	Male	18	FCT Abuja	Online	16	180,000	115,000	Part paid	Friend or fam
STO0012	19 Nov 2022	Web Development	Full Stack Web Development	SIWES	Undergraduate (University)	Female	23	Other	Physical	12	70,000	70,000	Paid in full	WhatsApp
STO0013	24 Nov 2022	Data Analysis	Python for Data Analysis	SIWES	Undergraduate (University)	Female	25	Nasarawa	Physical	12	70,000	40,000	Part paid	Friend or fam
STO0014	26 Nov 2022	Web Development	Frontend (HTML, CSS, JavaScript)	Regular	Graduate	Female	25	Kaduna	Online	10	100,000	45,000	Part paid	X (Twitter)
STO0015	03 Dec 2022	Data Analysis	SPSS for Research	Regular	Working professional	Female	26	FCT Abuja	Physical	6	60,000	60,000	Paid in full	Past student
STO0016	05 Dec 2022	Web Development	React	Regular	Graduate	Male	25	Kogi	Online	8	90,000	35,000	Part paid	X (Twitter)
STO0017	11 Dec 2022	Data Analysis	Python for Data Analysis	Regular	Undergraduate (University)	Male	24	FCT Abuja	Online	10	120,000	120,000	Paid in full	Instagram
STO0018	17 Dec 2022	Data Analysis	Full Data Analysis (Excel, SQL, Power BI)	Regular	Undergraduate (University)	Male	31	Kogi	Online	16	180,000	180,000	Paid in full	Friend or fam
STO0019	02 Jan 2023	Web Development	Frontend (HTML, CSS, JavaScript)	SIWES	Undergraduate (University)	Male	26	Kogi	Physical	24	120,000	80,000	Part paid	WhatsApp
STO0020	04 Jan 2023	Web Development	Frontend (HTML, CSS, JavaScript)	SIWES	Undergraduate (University)	Male	22	Niger	Physical	12	70,000	70,000	Paid in full	School place
STO0021	12 Jan 2023	Data Analysis	Excel and Power BI	SIWES	Undergraduate (University)	Male	24	FCT Abuja	Physical	24	120,000	120,000	Paid in full	School place
STO0022	12 Jan 2023	Web Development	Frontend (HTML, CSS, JavaScript)	Regular	Graduate	Female	23	Kogi	Physical	10	100,000	100,000	Paid in full	WhatsApp
STO0023	14 Jan 2023	Data Analysis	SQL for Data Analysis	Regular	Graduate	Male	21	FCT Abuja	Online	6	60,000	60,000	Paid in full	Instagram
STO0024	15 Jan 2023	Data Analysis	Python for Data Analysis	Regular	Working professional	Male	30	FCT Abuja	Online	10	120,000	75,000	Part paid	Friend or fam

The first rows of `tblEnrolments`.

5 · SOLUTION ARCHITECTURE

Table → PivotTables → formula layer → dashboards

```
Data (tblEnrolments)
   │
   ▼
Calc  ── 14 PivotTables on one shared PivotCache
   │       (each shaped for one chart or KPI)
   ▼
Model ── GETPIVOTDATA formulas wrapped in IFERROR
   │       (KPIs, percentages, chart ranges, dynamic captions)
   ▼
Dashboards ── charts, text boxes linked to Model cells, slicers
```

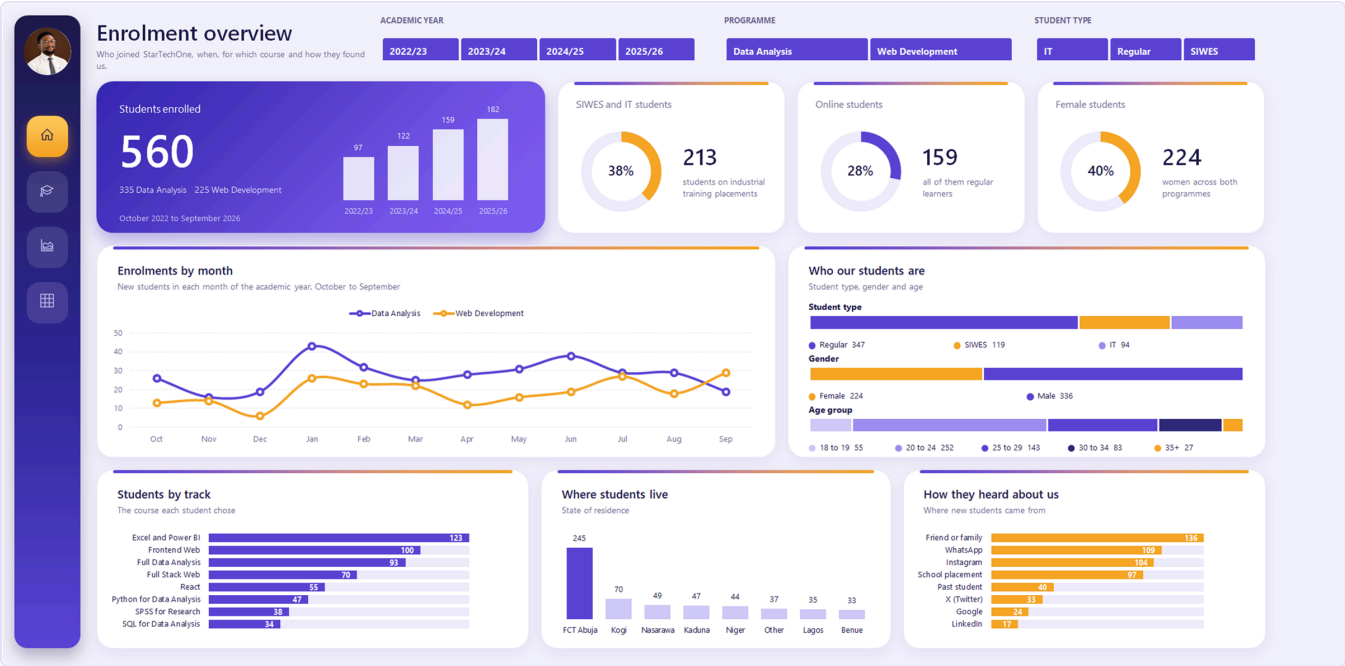
Why this design

- **One pivot cache, three slicers.** Academic year, Programme and Student type slicers connect to all 14 PivotTables and appear on both pages, so one click filters every KPI and chart.
- **A formula layer between pivots and charts.** Charts read stable ranges on the Model sheet, never a PivotTable directly. That avoids PivotChart field buttons, formatting resets and layouts that shift when a filter removes a row.
- `IFERROR(GETPIVOTDATA(...), 0)` **everywhere.** When a filter leaves no students (for example online SIWES), the dashboard shows zero rather than `#REF!`.
- **Dynamic text.** Headline numbers and captions are text boxes linked to Model cells, so they update with the slicers like the charts.
- **Ring charts.** Each ring is fed by `pct` and `MAX(0, 1 - pct)`, so it always sums to 100%.

How every number is calculated

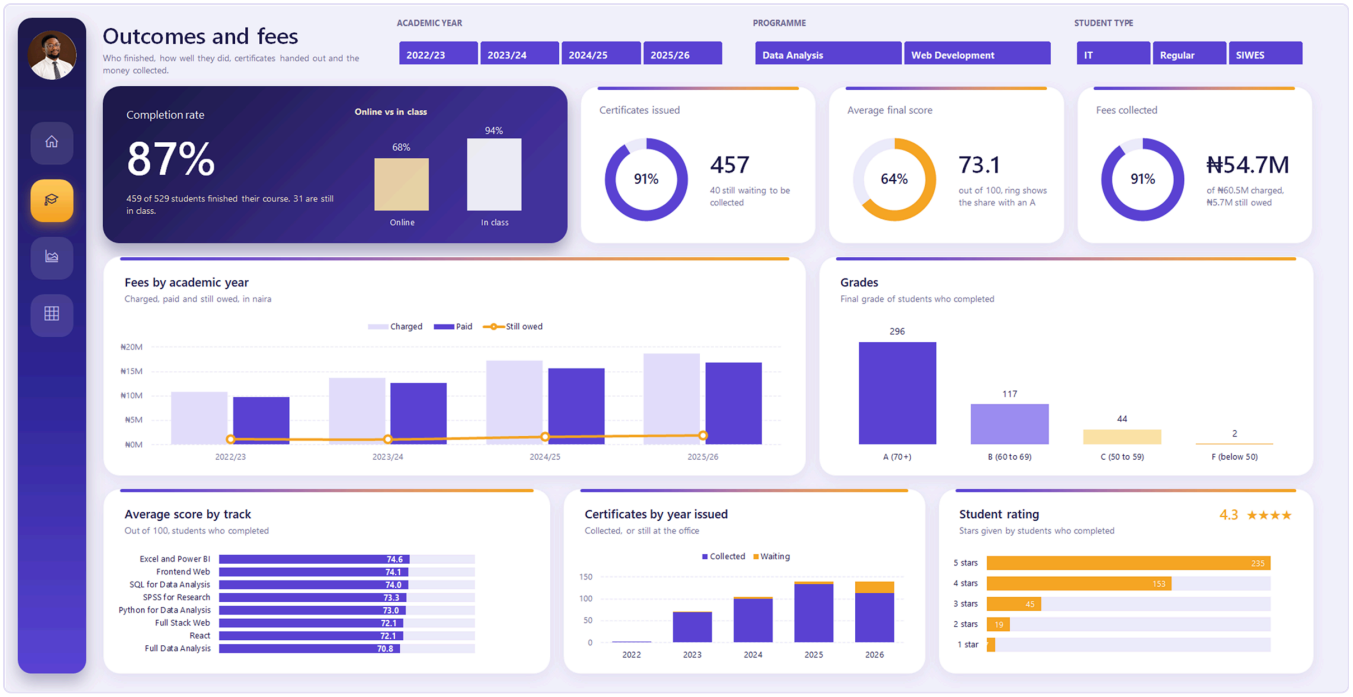
Metric	Definition
Students enrolled	Count of StudentID
SIWES and IT share	(SIWES + IT students) ÷ all students
Online share	LearningMode = Online ÷ all students
Female share	Female students ÷ all students
Completion rate	Completed = Yes ÷ (Yes + No); "In progress" left out
Certificates issued / waiting	Students with a certificate number / CertificateCollected = No
Average final score	Mean FinalScore of completers
Share with an A	Grade A ÷ completers
Fees charged / paid / owed	Sum of FeeNGN / AmountPaidNGN / OutstandingNGN
Fees collected %	Paid ÷ charged
Average rating	Mean 1 to 5 star rating of completers

Page 1: Enrolment overview



Section	What it shows
Hero KPI	560 students (335 Data Analysis, 225 Web Development) with intake per academic year
KPI rings	SIWES and IT students 38% (213), online 28% (159), female 40% (224)
Enrolments by month	One line per programme, October to September
Who our students are	100% stacked bars for student type, gender and age group
Track, state, channel	Students by track, state of residence, and how they heard about us

Page 2: Outcomes and fees



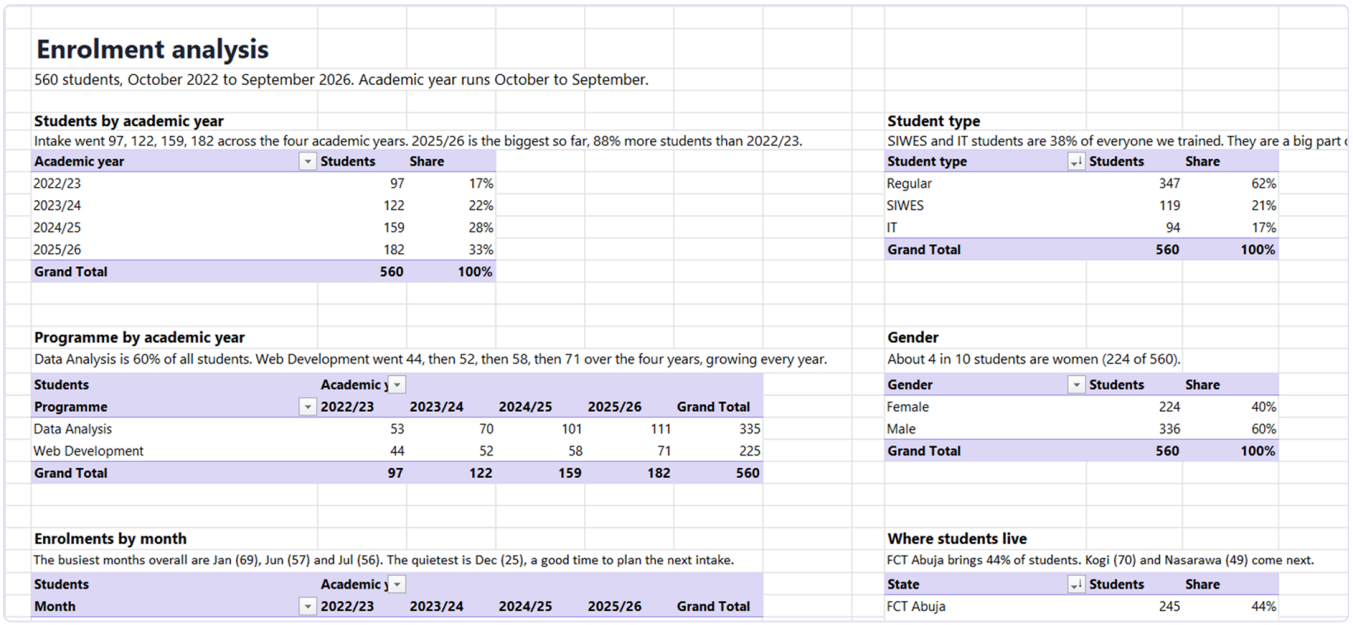
Section	What it shows
Hero KPI	87% completion, with online (68%) against in class (94%)
KPI rings	Certificates issued 457 (91% collected), average score 73.1 (64% with an A), fees collected ₦54.7M (91%)
Fees by academic year	Charged and paid as columns, still owed as a line
Grades, score by track	A to F counts for completers; average score by track, ranked
Certificates, rating	Certificates by year issued (collected vs waiting); 1 to 5 star distribution with the 4.3 average

Interactivity

- Three slicers filter every number on both pages.
- A navigation rail of buttons moves between the pages.
- KPI numbers and captions are linked text, so they never drift out of sync with the charts.

Supporting tables with a written takeaway above each

Two visible analysis sheets hold the PivotTables behind each chart, with one plain-English takeaway per table, so a reader can check any number on the dashboards.



Enrolment analysis

560 students, October 2022 to September 2026. Academic year runs October to September.

Students by academic year

Intake went 97, 122, 159, 182 across the four academic years. 2025/26 is the biggest so far, 88% more students than 2022/23.

Academic year	Students	Share
2022/23	97	17%
2023/24	122	22%
2024/25	159	28%
2025/26	182	33%
Grand Total	**560**	**100%**

Student type

SIWES and IT students are 38% of everyone we trained. They are a big part

Student type	Students	Share
Regular	347	62%
SIWES	119	21%
IT	94	17%
Grand Total	**560**	**100%**

Programme by academic year

Data Analysis is 60% of all students. Web Development went 44, then 52, then 58, then 71 over the four years, growing every year.

Students	Academic y				
Programme	2022/23	2023/24	2024/25	2025/26	Grand Total
Data Analysis	53	70	101	111	335
Web Development	44	52	58	71	225
Grand Total	**97**	**122**	**159**	**182**	**560**

Gender

About 4 in 10 students are women (224 of 560).

Gender	Students	Share
Female	224	40%
Male	336	60%
Grand Total	**560**	**100%**

Enrolments by month

The busiest months overall are Jan (69), Jun (57) and Jul (56). The quietest is Dec (25), a good time to plan the next intake.

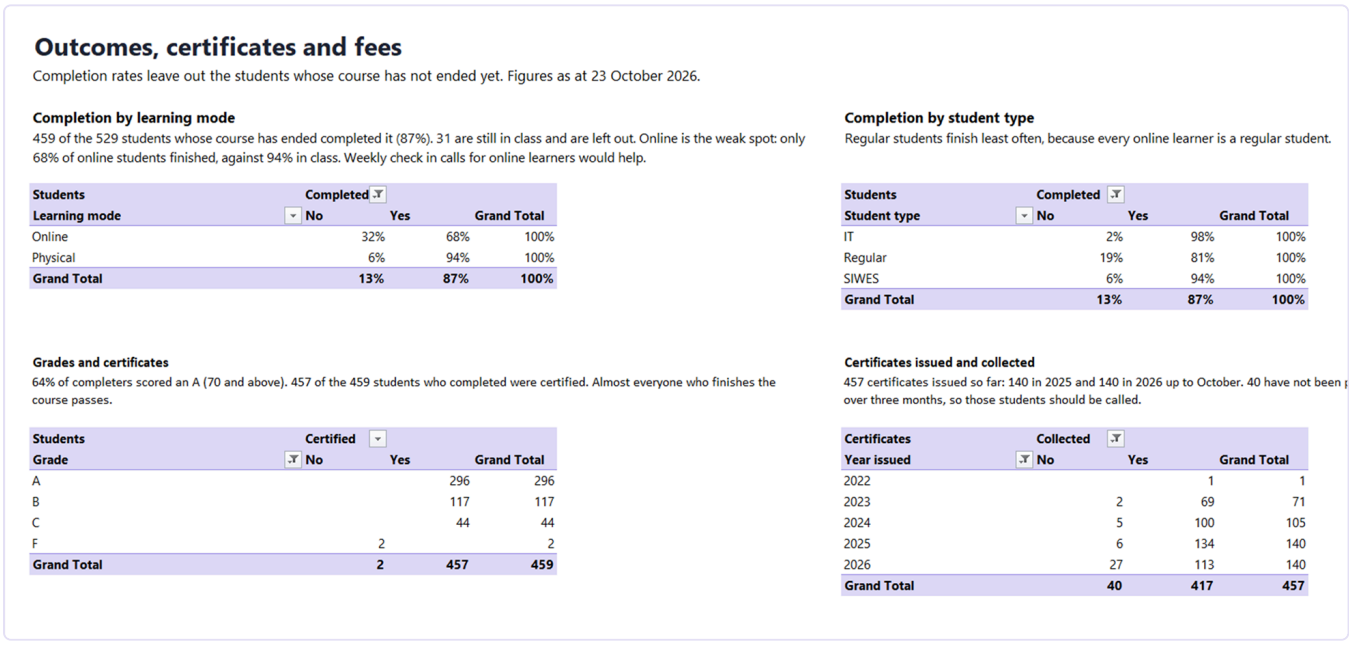
Students	Academic y				
Month	2022/23	2023/24	2024/25	2025/26	Grand Total

Where students live

FCT Abuja brings 44% of students. Kogi (70) and Nasarawa (49) come next.

State	Students	Share
FCT Abuja	245	44%

Enrolment analysis: students by year, programme by year, student type and gender.



Outcomes, certificates and fees

Completion rates leave out the students whose course has not ended yet. Figures as at 23 October 2026.

Completion by learning mode

459 of the 529 students whose course has ended completed it (87%). 31 are still in class and are left out. Online is the weak spot: only 68% of online students finished, against 94% in class. Weekly check in calls for online learners would help.

Students	Completed		
Learning mode	No	Yes	Grand Total
Online	32%	68%	100%
Physical	6%	94%	100%
Grand Total	**13%**	**87%**	**100%**

Completion by student type

Regular students finish least often, because every online learner is a regular student.

Students	Completed		
Student type	No	Yes	Grand Total
IT	2%	98%	100%
Regular	19%	81%	100%
SIWES	6%	94%	100%
Grand Total	**13%**	**87%**	**100%**

Grades and certificates

64% of completers scored an A (70 and above). 457 of the 459 students who completed were certified. Almost everyone who finishes the course passes.

Students	Certified		
Grade	No	Yes	Grand Total
A		296	296
B		117	117
C		44	44
F	2		2
Grand Total	**2**	**457**	**459**

Certificates issued and collected

457 certificates issued so far: 140 in 2025 and 140 in 2026 up to October. 40 have not been p over three months, so those students should be called.

Certificates	Collected		
Year issued	No	Yes	Grand Total
2022		1	1
2023	2	69	71
2024	5	100	105
2025	6	134	140
2026	27	113	140
Grand Total	**40**	**417**	**457**

Outcomes analysis: completion by mode and type, grades and certificates.

What the data says (no slicer applied)

Enrolment

1. **Intake has nearly doubled:** 97 (2022/23) to 182 (2025/26), +88%.
2. **Data Analysis leads, Web Development grows every year:** Data Analysis is 60% of students; Web Development went 44, 52, 58, 71.
3. **Placement students matter:** SIWES and IT are 38% of students; school placements alone brought in 97.
4. **Seasonality is clear:** January is busiest (69), then June (57) and July (56); December is quietest (25).
5. **Geography is concentrated:** FCT Abuja supplies 44% of students, then Kogi (70) and Nasarawa (49).
6. **Young cohort:** 55% are under 25; women are 40% of students.
7. **Channels differ by type:** for regular students Instagram (81) and friend or family (79) lead; friend or family is top overall (136).

Outcomes and fees

1. **Completion is 87%** (459 of 529 whose course has ended; 31 still in class).
2. **Online is the weak spot:** 68% of online students finish against 94% in class; Regular students complete at 81%, IT students at 98%.
3. **Attendance predicts dropout:** completers attended 88% of classes, dropouts 50%.
4. **Those who finish do well:** average score 73.1; 64% earned an A; 457 of 459 completers certified.
5. **Tracks are evenly matched:** 74.6 (Excel and Power BI) to 70.8 (Full Data Analysis).
6. **Satisfaction is high:** 51% gave 5 stars; average 4.3.
7. **Fees are 91% collected:** ₦60.5M charged, ₦54.7M paid, ₦5.7M owed; ₦3.3M of it sits with non-completers.
8. **Certificates are piling up:** 40 uncollected, 19 waiting over three months.

Seven actions, each backed by a number

#	Recommendation	Evidence
1	Add weekly check-in calls for online learners	Online completion 68% vs 94% in class
2	Use low attendance in the first weeks as an early-warning flag	Dropouts averaged 50% attendance vs 88%
3	Call the 19 students whose certificates have waited over three months	40 uncollected certificates
4	Protect and grow school placement partnerships	97 students from placements; SIWES and IT are 38%
5	Plan marketing and intake around the seasonal pattern	January peak, December low
6	Follow up part-paid fees before students drop out	₦3.8M owed by part-paid, ₦1.9M by not-paid students
7	Invest in Instagram and referral programmes for regular learners	Top two channels for regular students

Indigo, violet and gold on a card grid

Ink #1E1B4B titles	Primary #5B45D6 main series	Light #CFC8F8 tracks, context	Gold #F5A524 contrast, active

- **Gradients:** the sidebar runs from ink to violet; page 1's hero card is a violet gradient, page 2's a dark indigo one; every white card has a violet-to-gold line along its top edge.
- **One accent colour:** gold marks the second series, the active nav button and rating stars.
- **Slicers:** a custom style makes the selected option a solid violet pill, so the filter state is obvious.
- **Layout:** a fixed grid of 20 × 20 px cells; KPIs first, detail below; no gridlines or headers; zoomed so each page fits one screen.
- **Plain language:** every chart has a title and a one-line subtitle in plain English.

Open, filter, refresh, add students

1. Open the file in desktop Excel 2010 or later (Windows or Mac). Excel for the web shows slicers but has limited editing.
2. Start on **Enrolment Dashboard**; use the navigation rail to move to **Outcomes Dashboard**.
3. Click the Academic year, Programme or Student type pills to filter; click again or use the clear button to reset.
4. After editing the Data sheet, use **Data → Refresh All**; the Model sheet and charts update from the pivots.
5. To add students, type under the last row of `tblEnrolments`; the five calculated columns fill in automatically.

Data dictionary

Column	What it means	
About this data	Practice dataset of StarTechOne student enrolments, October 2022 to September 2026, 560 students. Figures are as at 23 October 2026. Made up for a portfolio dashboard; not real student records.	
StudentID	Unique code for each student (STO0001 and so on), in order of enrolment.	
EnrolmentDate	The day the student enrolled.	
Programme	Data Analysis or Web Development.	
Track	The course taken within the programme, for example Excel and Power BI, or Full Stack Web Development.	
StudentType	Regular (paying learner), SIWES (university students on industrial work experience) or IT (polytechnic students on industrial training).	
Status	Undergraduate, polytechnic student, graduate or working professional.	
Gender	Female or Male.	
Age	Age in years at enrolment.	
State	State the student lives in.	
LearningMode	Physical (in class) or Online. Only regular students can learn online; SIWES and IT students attend in person.	
DurationWeeks	Length of the course or the training placement in weeks.	
FeeNGN	Fee for the course in naira.	
AmountPaidNGN	How much the student actually paid.	
PaymentStatus	Paid in full, Part paid or Not paid.	
HeardAboutUs	How the student found StarTechOne, for example Instagram, WhatsApp or school placement.	
Attendance	Share of classes attended.	
Completed	Yes if the student finished the course, No if they dropped out, In progress if their course has not ended yet.	
FinalScore	Final assessment score out of 100. Blank if the student did not complete.	
Grade	A (70 and above), B (60 to 69), C (50 to 59), F (below 50). Blank if not completed.	
Certified	Yes if the student completed and scored 50 or more.	
Rating	The student's rating of the course, 1 to 5 stars. Blank if not completed.	
CourseEndDate	The day the course or training placement ends.	
CertificateNo	Certificate number, given in the order certificates were issued. Blank if no certificate.	
CertificateDate	The day the certificate was issued.	
CertificateCollected	Yes if the student has picked up the certificate, No if it is still with us.	

The Data Dictionary sheet defines every column and states that the data is synthetic.

What the numbers can and cannot tell you

Limitations

- The data is synthetic, so the findings illustrate the method and are not claims about a real school.
- Results are a snapshot as at 23 October 2026; there is no live data connection.
- Slicers and some formatting need desktop Excel.
- With narrow slicer selections, percentages rest on very few students and can swing widely.

Next steps

- Rebuild the model in Power BI with DAX measures for a side-by-side comparison.
- Add a cohort view from enrolment to certificate, and a dropout-risk view based on early attendance.
- Add a timeline slicer for rolling periods.
- Load the source with Power Query so refresh needs no manual pasting.

Prepared as a portfolio project by Taiwo Ajayi. All data is fictional.